CURRICULUM VITAE

CONTACT AND PERSONAL INFORMATION .

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PROFESSIONAL SUMMARY

Senior **AI Architect** with over 10 years of experience designing and leading **AI/ML, MLOps, and Generative AI** solutions across **Azure, AWS, and GCP**. Proven track record delivering **international, multi-country projects** (up to 15 countries) in **cybersecurity, telecom, energy, banking, and FMCG** sectors.
Experienced in supporting pre-sales and bid activities by defining AI architectures, assessing technical feasibility, and aligning solutions with business, compliance, and security requirements. Skilled at **bridging business and technical execution**, defining **product roadmaps**, managing **agile teams**, and delivering **MVPs** that scale to production. Expert in **GenAI (LLMs, RAG, agentic frameworks, LLMOps)** and **cloud-native data architectures**.

TECH STACK

* **Big Data & AI Ecosystem**: MongoDB, Hadoop, MapReduce, Hive, Pig, Impala, Flume, Sqoop, HBase, Apache Spark, Oozie, Solr, HCatalog, BigSheets, BigSQL, ML, DL, Python, Scala, R, RStudio, Visual Basic, Eclipse

* **Cloud & DevOps**: S3, Redshift, RDS, Glue, Lambda, EMR, Databricks, Data Factory, Synapse, Stream Analytics, Event Hub, Docker, Kubernetes, Redshift, BigQuery, Vertex AI, GCS, Cloud Run, SonarQube, Git, Maven, Terraform

* **Visualization & BI**: Tableau, QlikView, BigML, H2O

* **GenAI & Agents**: LangChain, LangGraph, LangSmith, AutoGen, OpenAI SDK, MCP, CrewAI, n8n, RAG

EMPLOYMENT HISTORY .

Indra – *AI Architect* (Sep 2024 – Present)

- **Project 1 – EUCINF (European Cyber Intelligence Framework)**
- Designed the end-to-end architecture of a large-scale ML platform on Google Cloud Platform (GCP) for ingesting and analyzing social network data to detect geopolitical trends.

- Defined a distributed ingestion layer using GCP Pub/Sub and Cloud Functions for processing high-volume streaming data in real-time.
- Implemented scalable data processing pipelines integrating BigQuery and Dataflow (Apache Beam) for advanced transformations and feature engineering.
- Designed ML lifecycle governance and model deployment using Vertex AI, including monitoring and explainability frameworks for sensitive intelligence data.
- Integrated NLP-based models for sentiment analysis and topic modeling, leveraging Vertex AI Search and Conversation (GenAI App Builder).
- Ensured GDPR compliance and data anonymization across multi-country deployments using Cloud DLP (Data Loss Prevention).
- Coordinated technical alignment across 10 European countries, defining common data contracts and interoperability standards.

- **Project 2 – AIDA (Autonomous Intelligent Defense Agents)**
- Designed a distributed multi-agent AI architecture for autonomous cyber threat detection and mitigation on AWS.
- Defined coordination protocols between AI agents using Amazon EventBridge and Amazon MQ for secure, event-driven communication.
- Designed UEBA-inspired behavioral models for detecting anomalous activity, utilizing Amazon SageMaker for model training and real-time inference.
- Integrated LLM-based reasoning components for contextual threat interpretation using Amazon Bedrock and agentic frameworks.
- Implemented MLOps and LLMOps pipelines for continuous model evaluation and secure deployment using AWS CodePipeline and SageMaker Pipelines.
- Designed scalable architecture across hybrid cloud (AWS/Azure) and secure defense environments (GovCloud standards).
- Coordinated with 15 international teams to ensure interoperability and standardized AI governance.

JTI (Japan Tobacco International) – *MLOps & Data Lake Development Manager* (Sep 2022 – Sep 2024)

- Led the design of enterprise-grade ML infrastructure in Azure Cloud for large-scale model deployment across global markets.
- Designed a modular Data Lake architecture aligned with Data Mesh principles, enabling domain-oriented data ownership.
- Implemented CI/CD pipelines for ML models using Azure DevOps, Docker, and Infrastructure as Code (Terraform).
- Designed model registry, monitoring, drift detection, and retraining frameworks using MLflow and Azure ML.
- Led Generative AI initiatives including:
- Internal knowledge assistants using RAG pipelines
- LLM-powered document search and summarization tools
- Prompt engineering standards and governance frameworks
- Implemented secure data access models using RBAC and managed identities.

- Defined LLMOps practices including prompt versioning, evaluation datasets, guardrails, and cost optimization strategies.
- Collaborated with data teams from 5 countries, aligning data standards and ML deployment patterns globally.
- Managed cross-functional teams including Data Engineers, ML Engineers, and Cloud Architects.

Babel – *Big Data Architect* (Apr 2020 – Sep 2022)

- **Project 1 - Orange (Telecom Sector)**
- Designed large-scale data preparation pipelines for AI-driven customer analytics and churn prediction.
- Implemented Spark-based transformation workflows in AWS (EMR, S3, Glue).
- Developed optimized feature engineering pipelines in Scala.
- Designed batch and near real-time ingestion processes integrating telecom event streams.
- Improved data quality frameworks and validation processes across distributed datasets.
- **Project 2 - Canal Isabel II (Utilities Sector)**
- Designed Azure cloud architecture using Infrastructure as Code (Terraform).
- Implemented a scalable Data Platform integrating Data Factory, Databricks, and Synapse.
- Developed ingestion pipelines from heterogeneous sources (IoT sensors, ERP systems, relational databases).
- Designed monitoring and observability frameworks for pipeline reliability.
- Contributed to digital transformation roadmap including AI use cases for predictive maintenance and demand forecasting.

Paradigma Digital – *Big Data Architect* (Sep 2018 – Apr 2020)

- **Gas Natural:** Defined **Azure cloud architecture**, implemented ingestion pipelines using **Blob Storage, Azure Functions, and SQL Database**.
- **Iberdrola:** Deployed AI models in Azure; designed data and software **quality frameworks**.
- **Mutua Madrileña:** Designed a **metadata-based scheduler** for the data lake; optimized data ingestion and transformation processes; contributed to roadmap planning.

Banco Popular – *Big Data Engineer / Data Scientist* (Jun 2017 – Sep 2018)

- Developed **machine learning models** (R, Shiny, Python, ARIMA) within **Cloudera**.
- Delivered analytical solutions for the **Advanced Analytics Department** using **QlikView, Hive, and Spark**.

Everis (now NTT Data) – *Big Data Architect* (Sep 2015 – Mar 2017)

- Worked for **Banco Santander** using **Cloudera distribution**.
- Designed and implemented **data lake transformations**, including real-time and batch updates using **Kafka, HBase, Scala, Hive, Hadoop, and Spark Streaming**.

EDUCATION

* **Degree in Electronic and Automation Engineering**, University of the Basque Country (Spain), 1987–1992; University of Antonio Nebrija, 2011

* **Degree in Business Administration**, UNED (Open University), 2008–2012

* **Master's Degree in Stock Markets and Financial Derivatives**, UNED (Open University), 2009–2010 — Graduated with distinction

* **Master in Cloud Computing (Azure, AWS, Google Cloud Platform)**, CICE, 2020

* **Master's Degree in Big Data and Business Intelligence**, CICE, Madrid, 2015

POSTGRADUATE COURSES AND CERTIFICATIONS

* **Certification AWS Certified Data Analytics – Specialty.**
* **Certification AZ-303, Microsoft Azure Architect Technologies.**
* **Certification AZ-304, Microsoft Azure Architect Design.**
* **Certification DP-203 Microsoft Azure Data Engineer.**
* **Certification Google Cloud Data Engineer.**
* **Certification Generative AI Fundamentals.**
* **Course of Machine Learning,** imparted by Stanford University in 2015, with a length of 80 hours.
* **Course of IBM Big Data (InfoSphere BigInsight)**, imparted by IBM University in 2015, with a length of 160 hours.
* **Course of Big Data Architecture,** imparted by Formación Hadoop in 2015, with a length of 50 hours.
* **Course of Big Data with Apache Spark,** imparted by BerkeleyX University in 2015, with a length of 50 hours.
* **Course of AWS Cloud Solution Architect**, imparted by A Cloud Guru in 2016.
* **MCP Masterclass: Complete Guide to MCP in Python,** Udemy, 2025
* **The Complete Agentic AI Engineering Course**, Udemy, 2025
* **AI Automation: Build LLM Apps and AI-Agents with n8n and APIs**, Udemy, 2025
* **LLM Engineering: Master AI, Large Language Models & Agents, Udemy, 2025**

LENGUAGES

- **English**: Upper-intermediate (Cambridge First Certificate, B2); current proficiency estimated at C1 due to frequent use
- **Spanish: Native / bilingual**

PERSONAL PROJECTS

Custom Web Chatbot using n8n and LLMs (2025)

- Developed a fully functional AI chatbot integrated into websites using **n8n** workflows, **OpenAI API**, and **custom HTML embedding**.
- Implemented **RAG (Retrieval-Augmented Generation)** pipelines for context retrieval and **multi-turn conversation memory**.
- **Designed scalable deployment using Docker and API endpoints for easy integration into multiple sites.**

AI-Generated Portfolio Website (2026**)**

Designed and orchestrated a multi-agent AI system using CrewAI to collaboratively generate my portfolio website:

- Designed and orchestrated a multi-agent AI system using CrewAI to collaboratively generate this portfolio website
- Three specialized agents — Web Designer, Frontend Developer, and Quality Reviewer — working in sequence to plan, build, and refine the final output
- Knowledge-driven generation: agents read structured CV data and user-driven preferences to produce personalized, production-ready HTML/CSS
- Full project architecture designed from scratch, including agent roles, task pipelines, and output validation